

Nivell 1 - Exercici 1

A partir dels documents adjunts (estructura_dades i dades_introduir), importa les dues taules. Mostra les característiques principals de l'esquema creat i explica les diferents taules i variables que existeixen. Assegura't d'incloure un diagrama que il·lustri la relació entre les diferents taules i variables.

1.1. - He creado a base de datos y las 2 tablas y la relación entre ellas

```
14 • CREATE TABLE IF NOT EXISTS company (  
15     id VARCHAR(15) NOT NULL,  
16     company_name VARCHAR(255),  
17     phone VARCHAR(15),  
18     email VARCHAR(100),  
19     country VARCHAR(100),  
20     website VARCHAR(255),  
21     PRIMARY KEY(id)  
22 );  
23
```

```
24 • CREATE TABLE IF NOT EXISTS transaction (  
25     id VARCHAR(255) NOT NULL,
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

Table	Create Table
company	CREATE TABLE `company` (`id` varchar(15) ...

Result 1 x

Output

Action Output

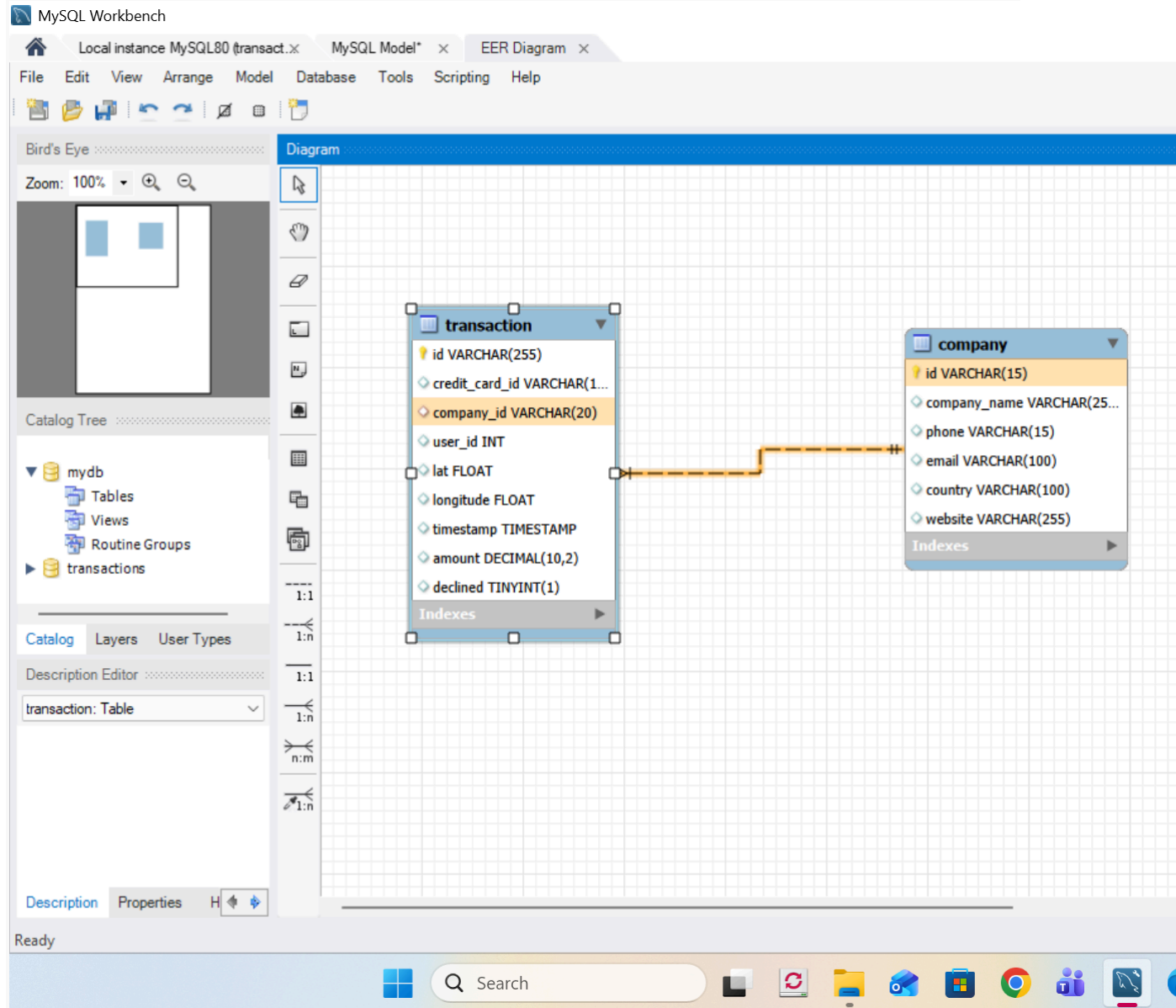
#	Time	Action	Message
4	01:09:32	CREATE TABLE IF NOT EXISTS company (id VARCHAR(15), company_name VARCHAR(255)	Error Code: 1064. You have an error in your SQL syntax; check th
5	01:10:42	CREATE TABLE IF NOT EXISTS company (id VARCHAR(15), company_name VARCHAR(255)	0 row(s) affected, 1 warning(s): 1050 Table 'company' already exists
6	01:36:02	SHOW CREATE TABLE company	1 row(s) returned

1.2 - Características de l'esquema

- La eschema TRANSACTIONS está compuesta por 2 tablas que se relacionan a través del código de identificación de empresa (company id).
- A cada id de la tabla “company” le corresponden varias transacciones de la tabla “transaction”.
- La tabla “transaction” es la tabla de hechos: es un registro de (los intentos de) transacciones con tarjeta de crédito hechas por usuarios de diferentes empresas.
- En la tabla de hechos encontramos variables como id de la transacción, usuario, id de empresa, importe, tarjeta de crédito, coordenadas geográficas de la ubicación del usuario, hora.
- También aparece la columna “declined” que valida si la transacción falló o finalizó con éxito, según si el pago fue denegado o no.
- La tabla “company” es una tabla de dimensiones. En ella aparecen el nombre de la empresa y detalles de contacto como email, teléfono, página web.

He añadido una relación de 1 a muchos entre la tabla “company” y la tabla “transaction”: cada company puede aparecer en muchas transacciones. Lo he hecho via código (CONSTRAINT en la creacion de la tabla “transaction”) y estableciendo el número de identificación de empresa como clave foránea (foreign key).

1.3 - Diagrama



Nivell 1 - Exercici 2

Utilitzant JOIN realitzaràs les següents consultes:

2.1 - Llistat dels països que estan generant vendes.

The screenshot shows a SQL IDE interface with a query editor and a results pane. The query editor contains the following SQL code:

```
0154 # Llistat dels països que estan generant vendes.
0155 # Des de quants països es generen les vendes.
0156 # Identifica la companyia amb la mitjana més gran de vendes.
0157
0158 SELECT DISTINCT country
0159 FROM transactions.company
0160 JOIN transactions.transaction
0161 ON transaction.company_id = company.id
0162 WHERE declined = 0;
0163
```

The results pane shows the output of the query, which is a list of countries:

country
Netherlands
Sweden
Ireland
United States
Belgium

Below the results pane, there is an "Action Output" section showing a log of database actions and their messages:

#	Time	Action	Message
100121	18:29:20	ALTER TABLE transaction DROP CONSTRAINT IF EXISTS transaction_ibfk_1	Error Code: 1064. You have an error in your SQL syntax; check the manual that co
100122	18:32:23	ALTER TABLE transaction DROP CONSTRAINT fk_company	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
100123	18:34:28	ADD CONSTRAINT fk_company FOREIGN KEY (company_id) REFERENCES company(id)	Error Code: 1064. You have an error in your SQL syntax; check the manual that co
100124	18:34:49	ALTER TABLE transaction ADD CONSTRAINT fk_company FOREIGN KEY (company_id) REFERENC...	100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0
100125	18:46:58	SELECT DISTINCT country FROM transactions.company JOIN transactions.transaction ON transaction....	15 row(s) returned

2.2 - Des de quants països es generen les vendes.

```
55
56 # 2.2 - Des de quants països es generen les vendes.
57
58 • SELECT COUNT(DISTINCT country) AS Num_paises
59 FROM transactions.company
60 JOIN transactions.transaction
61 ON transaction.company_id = company.id
62 WHERE declined=0;
63
64 # 2.3 - Identifica la companyia amb la mitjana més gran de vendes.
65
66 • SELECT ROUND(AVG(amount),2) AS Media_ventas, company_name
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Num_paises
15

Result 5 x

Output

Action Output

#	Time	Action	Message
13	06:15:34	SELECT COUNT (DISTINCTcountry) AS Num_paises FROM transactions.company JOIN transactions.tr...	Error Code: 1630. FUNCTION transa
14	06:16:00	SELECT COUNT (DISTINCT country) AS Num_paises FROM transactions.company JOIN transactions.tr...	Error Code: 1064. You have an error
15	06:16:58	SELECT COUNT(DISTINCT country) AS Num_paises FROM transactions.company JOIN transactions.tr...	1 row(s) returned

2.3 - Identifica la companyia amb la mitjana més gran de vendes.

```
# 2.3 - Identifica la companyia amb la mitjana més gran de vendes.
SELECT ROUND(AVG(amount),2) AS Media_ventas, company_name
FROM transactions.company
JOIN transactions.transaction
ON transaction.company_id = company.id
WHERE declined = 0
GROUP BY company_name
ORDER BY AVG(amount) DESC
LIMIT 1;

# Nivell 1 - Exercici 3
```

Result Grid

Media_ventas	company_name
284.91	Ac Fermentum Incorporated

Result 3

#	Time	Action	Message	Duration / F
6	01:36:02	SHOW CREATE TABLE company	1 row(s) returned	0.000 sec /
7	02:08:29	SELECT ROUND(AVG(amount),2) AS Media_ventas, company_name FROM transactions.company JOIN...	101 row(s) returned	1.000 sec /
8	02:09:17	SELECT ROUND(AVG(amount),2) AS Media_ventas, company_name FROM transactions.company JOIN...	1 row(s) returned	0.906 sec /

Nivell 1 - Exercici 3

Utilitzant només subconsultes (sense utilitzar JOIN):

3.1 - Mostra totes les transaccions realitzades per empreses d'Alemanya.

```
%0183      # Exercici 3
%0184
%0185      SELECT *
%0186      FROM transaction
%0187      WHERE company_id IN (SELECT id FROM company WHERE country = "Germany")
%0188      AND declined = 0
%0189      AND id is not Null;
%0190
%0191
%0192
%0193
%0194
%0195
```

Result Grid										
Filter Rows:										
Edit:										
Export/Import:										
Wrap Cell Content:										
Fetch rows:										
	id	credit_card_id	company_id	user_id	lat	longitude	timestamp	amount	declined	
▶	00138D3B-206D-4C03-94B7-63A2676EB9B4	CcS-4899	b-2222	318	41.3781	12.447	2020-03-25 10:43:43	426.36	0	
	0013C1B6-3B84-4D6C-8154-E2B3FEBCA8E9	CcS-5070	b-2222	489	41.3814	2.18176	2020-12-17 18:15:37	316.90	0	
	00201A11-2E62-44C4-941D-198FC8DB77F0	CcU-3512	b-2222	193	55.5704	-3.65129	2021-01-22 23:44:27	453.04	0	
	00235618-0A5C-4D49-9DCB-B3A9405D8923	CcS-8137	b-2222	3556	59.8421	18.729	2020-09-09 15:43:19	263.14	0	
	005A5A7B-1F1A-4B6C-9B15-1625A78C9C38	CcS-8998	b-2222	4417	41.1591	-8.63905	2024-05-15 09:10:11	442.01	0	

transaction 5 x

Output

Action Output

#	Time	Action	Message
✓ 4	20:49:23	SELECT * FROM transaction WHERE company_id IN (SELECT id FROM company WHERE country = "...	13269 row(s) returned
✓ 5	20:49:42	SELECT * FROM transaction WHERE company_id IN (SELECT id FROM company WHERE country = "...	13269 row(s) returned

3.2 - Llista les empreses que han realitzat transaccions per un amount superior a la mitjana de totes les transaccions.

He a adido ALIAS + declined=0

The screenshot shows a SQL IDE interface. The top pane contains a SQL query with line numbers 80 to 90. The query is as follows:

```
80 FROM transaction
81 WHERE company_id IN (SELECT id FROM company WHERE country = "Germany")
82 AND declined = 0;
83
84 # 3.2 - Llista les empreses que han realitzat transaccions per un amount superior a la mitjana de totes les transaccions.
85
86 • SELECT company_id, (SELECT company_name FROM company t WHERE t.id = company_id )
87 FROM transaction
88 WHERE amount > (SELECT AVG(amount) AS Mitjana FROM transaction)
89 AND declined = 0
90 GROUP BY company_id;
```

The bottom pane shows the 'Result Grid' with a table of results. The table has two columns: 'company_id' and '(SELECT company_name FROM company t WHERE t.id = company_id)'. The results are as follows:

company_id	(SELECT company_name FROM company t WHERE t.id = company_id)
b-2222	Ac Fermentum Incorporated
b-2226	Magna A Neque Industries
b-2230	Fusce Corp.
b-2234	Convallis In Incorporated

Below the result grid, the 'Output' pane shows a log of actions. The log includes the following entries:

#	Time	Action	Message	Duration / Fetch
22	15:55:32	CREATE TABLE credit_card_status AS SELECT transactions.credit_card_id AS credit_card_id, CAS...	Error Code: 1054. Unknown column 'transactions.credit_card_id' in 'field list'	0.000 sec
23	16:10:59	SELECT company_id, (SELECT company_name FROM company t WHERE t.id = company_id) FROM tr...	Error Code: 1146. Table 'inmasedatabase.transaction' doesn't exist	0.016 sec
24	16:11:13	SELECT company_id, (SELECT company_name FROM company t WHERE t.id = company_id) FROM tr...	100 row(s) returned	0.985 sec / 0.000 sec

3.3 - Eliminaran del sistema les empreses que no tenen transaccions registrades, entrega el llistat d'aquestes empreses.

```
90
91 # 3.3 - Eliminaran del sistema les empreses que no tenen transaccions registrades, entrega el llistat d'aquestes empreses.
92
93 SELECT id, company_name
94 FROM company AS c
95 WHERE id NOT IN (SELECT company_id FROM transaction AS t WHERE c.id = t.company_id);
96
97 DELETE FROM company
98 WHERE id NOT IN (SELECT company_id FROM transaction AS t WHERE company.id = t.company_id);
99
100
101
102
103
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	11:02:53	DELETE FROM company WHERE id NOT IN (SELECT company_id FROM transaction AS t WHERE c.i...	Error Code: 1054. Unknown column 'c.id' in 'where clause'	0.000 sec
3	11:03:31	DELETE FROM company WHERE id NOT IN (SELECT company_id FROM transaction AS t WHERE co...	0 row(s) affected	0.016 sec
4	11:03:53	DELETE FROM company WHERE id NOT IN (SELECT company_id FROM transaction AS t WHERE co...	0 row(s) affected	0.016 sec

Exercici 4

4.1 - La teva tasca és dissenyar i crear una taula anomenada "credit_card" que emmagatzemi detalls crucials sobre les targetes de crèdit. La nova taula ha de ser capaç d'identificar de manera única cada targeta i establir una relació adequada amb les altres dues taules ("transaction" i "company").

The screenshot shows a database management tool interface. On the left, a tree view displays the database structure, including tables like 'company', 'credit_card', and 'transaction'. The 'credit_card' table is selected, and its columns are listed: 'id' (varchar(60) PK), 'iban' (varchar(60)), 'pan' (varchar(60)), 'pin' (varchar(60)), 'cvv' (varchar(60)), and 'expiring_date' (varchar(60)).

The main area displays the SQL code for creating the 'credit_card' table:

```
# Exercici 4
CREATE TABLE IF NOT EXISTS credit_card (
  id VARCHAR(60) PRIMARY KEY,
  iban VARCHAR(60),
  pan VARCHAR(60),
  pin VARCHAR(60),
  cvv VARCHAR(60),
  expiring_date VARCHAR(60)
);
```

The bottom section shows the 'Output' window, which displays the results of the SQL execution. The output is a table with columns: #, Time, Action, and Message.

#	Time	Action	Message
6	20:35:59	SELECT * FROM transaction WHERE company_id IN (SELECT id FROM comp...	13269 row(s) returned
7	20:37:08	SELECT * FROM transaction WHERE company_id IN (SELECT id FROM comp...	13291 row(s) returned
8	20:37:37	SELECT * FROM transaction WHERE company_id IN (SELECT id FROM comp...	13291 row(s) returned
9	20:40:43	SELECT company_id, amount FROM transaction WHERE amount > (SELECT ...	Error Code: 1064. You have an error in your SQL syntax; check the manual that ...
10	20:42:01	SELECT company_id, amount FROM transaction WHERE amount > (SELECT ...	46722 row(s) returned
11	20:43:26	SELECT company_id, amount FROM transaction WHERE amount > (SELECT ...	46722 row(s) returned
12	20:46:15	SELECT company_name FROM company WHERE id IN (SELECT company_id ...	0 row(s) returned
13	20:48:00	DELETE FROM company WHERE id IN (SELECT company_id FROM transacti...	0 row(s) affected
14	20:53:29	CREATE TABLE IF NOT EXISTS credit_card (id VARCHAR(60) PRIMAR...	0 row(s) affected

Propuesta de mejora de la creación de tabl inicial (ajuste tipo de datos):

```
CREATE TABLE IF NOT EXISTS credit_card (
```

```
    id VARCHAR(8) PRIMARY KEY,
```

```
    iban VARCHAR(24),
```

```
    pan VARCHAR(60),
```

```
    pin INT(4),
```

```
    cvv INT(3),
```

```
    expiring_date DATE
```

```
);
```


4.2 - Després de crear la taula serà necessari que ingressis la informació del document denominat "dades_introduir_credit". Recorda mostrar el diagrama i realitzar una breu descripció d'aquest.

CHEMAS

Filter objects

- gpt_filtre_agr
- gpt_join_subq
- hospitales
- irinasdatabase
- olympics
- sakila
- sys
- tienda_online
- transactions**
 - Tables
 - company
 - credit_card**
 - Columns
 - Indexes
 - Foreign Keys
 - Triggers
 - transaction
 - Views

Administration Schemas

Information

Table: credit_card

Columns:

- id varchar(60) PK
- iban varchar(60)
- pan varchar(60)
- pin varchar(60)
- cvv varchar(60)
- expiring_date varchar(60)

Object Info Session

1 • **SELECT * FROM transactions.credit_card;**

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows

	id	iban	pan	pin	cvv	expiring_date
▶	CcS-4857	XX4857591835292505850771	2314242385113924	1819	467	09/27/25
	CcS-4858	XX8581768137002436094025	6582720299715533	3964	817	12/28/28
	CcS-4859	XX7826930491423553609370	8861684536289642	4983	277	11/26/26
	CcS-4860	XX5559590368835304645299	2481155515498459	6876	661	07/27/27
	CcS-4861	XX2035182877195191627307	1308930301149557	5710	398	04/25/26

credit_card 1 x

Output

Action Output

#	Time	Action	Message
✓ 5027	21:05:55	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS...	1 row(s) affected
✓ 5028	21:06:06	SELECT * FROM transactions.credit_card	5000 row(s) returned

er objects

gpt_filtre_agr
gpt_join_subq
hospitales
irinasdatabase
olympics
sakila
sys
tienda_online

transactions

Tables

company

credit_card

Columns

Indexes

Foreign Keys

Triggers

transaction

Views

stration Schemas

ation

ole: credit_card

lums:

d varchar(60)
PK
ban varchar(60)
pan varchar(60)
pin varchar(60)
cvv varchar(60)
expiring_date varchar(60)

Info Session

ompleted

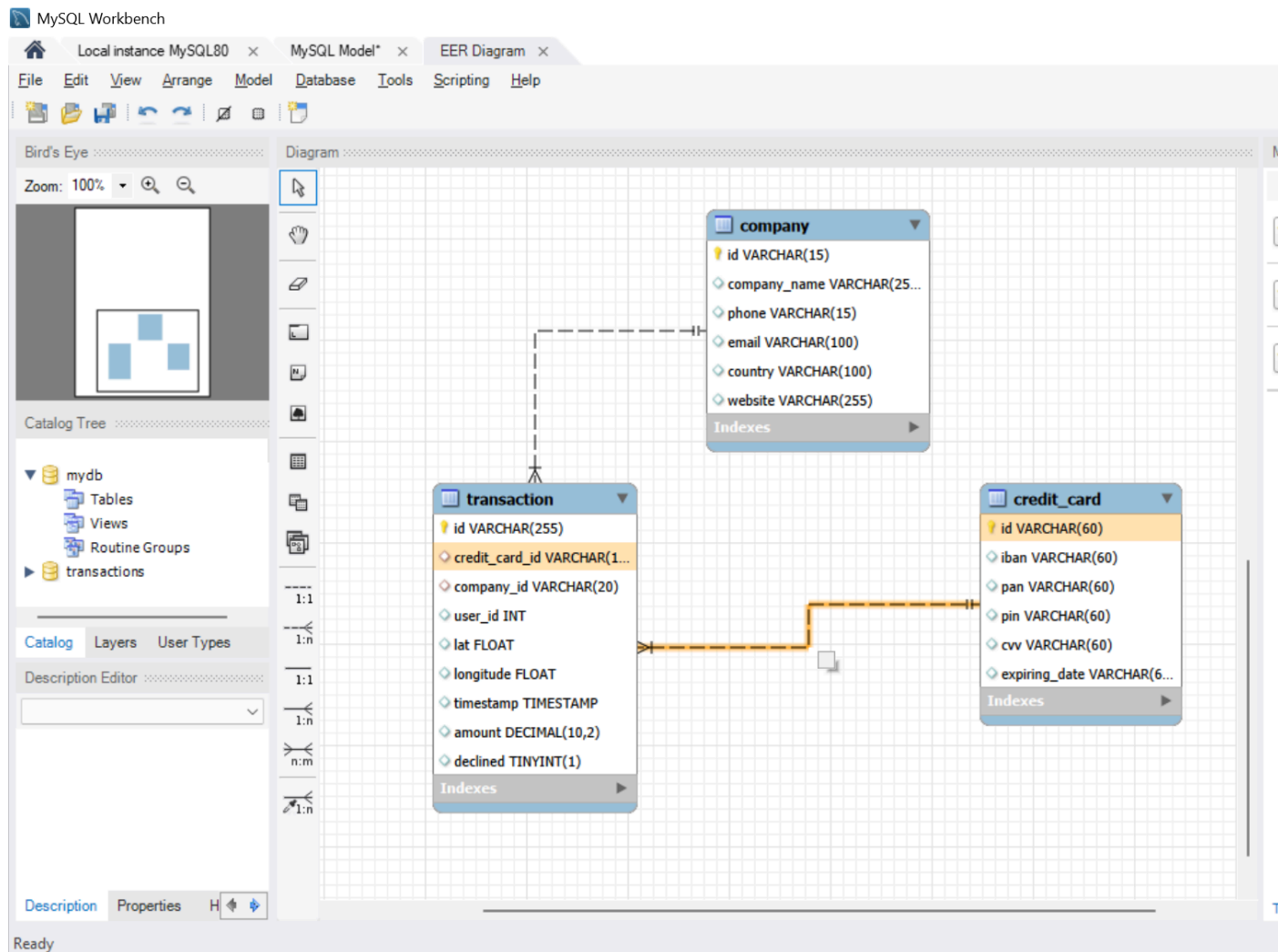
```
5191 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9563', 'XX399  
5192 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9564', 'XX64  
5193 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9565', 'XX64  
5194 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9566', 'XX32  
5195 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9567', 'XX85  
5196 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9568', 'XX38  
5197 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9569', 'XX0  
5198 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9570', 'XX63  
5199 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9571', 'XX48  
5200 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9572', 'XX40  
5201 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9573', 'XX96  
5202 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9574', 'XX66  
5203 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9575', 'XX49  
5204 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9576', 'XX52  
5205 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9577', 'XX15  
5206 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9578', 'XX99  
5207 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9579', 'XX29  
5208 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9580', 'XX78  
5209 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9581', 'XX91  
5210  
5211
```

Output

Action Output

#	Time	Action	Message
5027	21:05:55	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS...	1 row(s) affected
5028	21:06:06	SELECT * FROM transactions.credit_card	5000 row(s) returned

Nuestro modelo estrella tiene una nueva tabla de dimensiones: `credit_card`. Para añadirla al modelo, he creado una relación de 1 a muchos con la tabla de hechos (“`transaction`”) a través del número de identificación de la tarjeta de crédito (`credit_card_id` / `id`).



Exercici 5

El departament de Recursos Humans ha identificat un error en el número de compte associat a la targeta de crèdit amb ID CcU-2938. La informació que ha de mostrar-se per a aquest registre és: TR323456312213576817699999. Recorda mostrar que el canvi es va realitzar.

```
210
217   # Exercici 5
218
219 • UPDATE credit_card
220 SET iban = "TR323456312213576817699999"
221 WHERE id = "CcU-2938";
222
223
224
225
226
227
228
229
230
```

Output

Action Output

#	Time	Action	Message
5038	21:23:18	UPDATE credit_card SET iban "TR323456312213576817699999" WHERE id = "CcU-2938"	Error Code: 1064. You have an error in your SQL syntax; check the
5039	21:23:29	UPDATE credit card SET iban = "TR323456312213576817699999" WHERE id = "CcU-2938"	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

Query Completed

ANTES

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Irina_Mihai_Tasca S2.01. Nocio... credit_card x

Don't Limit

1 • `SELECT * FROM transactions.credit_card;`

Result Grid

Filter Rows: CcU-2938

Edit: Export/Import: Wrap Cell Content: Fetch rows:

	id	iban	pan	pin	cvv	expiring_date
▶	CcU-2938	TR301950312213576817638661	5424465566813633	3257	984	10/30/22
*	NULL	NULL	NULL	NULL	NULL	NULL

credit_card 1 x

Output

Action Output

#	Time	Action	Message	Duration
5029	21:11:45	ALTER TABLE credit_card ADD CONSTRAINT fk_card FOREIGN KEY (id) REFERENCES transaction(c...	Error Code: 1822. Failed to add the foreign key constraint. Missing index for constraint 'fk_card' in the refer...	0.000 s
5030	21:15:12	ALTER TABLE transaction ADD CONSTRAINT fk_card FOREIGN KEY (credit_card id) REFERENCES ...	100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0	2.485 s

Ready

DESPUÉS

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Irina_Mihai_Tasca S2.01. Nocio... credit_card x

Don't Limit

```
1 • SELECT * FROM transactions.credit_card;
```

Result Grid

Filter Rows: CcU-2938

Edit: Export/Import: Wrap Cell Content: Fetch rows:

	id	iban	pan	pin	cvv	expiring_date
▶	CcU-2938	TR323456312213576817699999	5424465566813633	3257	984	10/30/22
*	NULL	NULL	NULL	NULL	NULL	NULL

credit_card 1 x

Output

Action Output

#	Time	Action	Message
5038	21:23:18	UPDATE credit_card SET iban = "TR323456312213576817699999" WHERE id = "CcU-2938"	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'TR323456312213576817699999' at line 1
5039	21:23:29	UPDATE credit card SET iban = "TR323456312213576817699999" WHERE id = "CcU-2938"	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

Query Completed

Exercici 6

En la taula "transaction" ingressa una nova transacció amb la següent informació:

Id

108B1D1D-5B23-A76C-55EF-C568E49A99DD

credit_card_id

CcU-9999

company_id

b-9999

user_id

9999

lat

829.999

longitude

-117.999

amount

111.11

declined

0

Debido a que las tablas están relacionadas entre ellas, he tenido que crear primero un registro para la empresa nueva y otro registro para la nueva tarjeta en las tablas relacionadas “company” respectivamente “credit_card”. Solo después he podido insertar la nueva transacción.

```
5222
5223 # Exercici 6
5224
5225
5226 • INSERT INTO credit_card (id) VALUES ("CcU-9999");
5227 • INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined) VALUES ("10881D1D-5B23-A"
5228 "b-9999", "9999", "829.999", "-117.999", "111.11", "0");
5229
5230
5231
5232
5233
5234
5235
5236
```

Output

#	Time	Action	Message
5045	21:35:53	INSERT INTO credit_card (id) VALUES ("CcU-9999")	1 row(s) affected
5046	21:35:53	INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined) VAL...	1 row(s) affected

Query Completed

La nueva fila se ve aquí:

Don't Limit

1 • SELECT * FROM transactions.transaction;

Result Grid

Filter Rows: :F-C568E49A99DD

Edit:

Export/Import:

Wrap Cell Content:

Fetch rows:

id	credit_card_id	company_id	user_id	lat	longitude	timestamp	amount	decline
108B1D1D-5B23-A76C-55EF-C568E49A99DD	CcU-9999	b-9999	9999	829.999	-117.999	NULL	111.11	0
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

transaction 1 x

Output

Action Output

#	Time	Action	Message
5046	21:35:53	INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitud...	1 row(s) affected
5047	21:38:20	SELECT * FROM transactions.transaction	100001 row(s) returned

Para la compañía b-9999 se ha creado una fila nueva en la tabla company con la única información disponible: el id de la company.

El código utilizado:

INSERT INTO company (id) VALUES ("b-9999");

The screenshot shows the SQL Developer interface. On the left, the 'HEMAS' schema is expanded, showing the 'company' table. The table's columns are listed: id, company_name, phone, email, country, and website. The 'id' column is marked as the primary key (PK). Below the table list, the 'Table: company' section shows the column data types: id is varchar(15), company_name is varchar(255), and phone is varchar(15). In the center, a query is entered: `SELECT * FROM transactions.company;`. Below the query, the 'Result Grid' shows the results of the query, filtered for 'b-9999'. The result grid has columns: id, company_name, phone, email, country, and website. The first row shows the value 'B-9999' in the 'id' column, with all other columns containing 'NULL'.

Exercici 7

Des de recursos humans et sol·liciten eliminar la columna "pan" de la taula credit_card. Recordar mostrar el canvi realitzat.

```
35230
35231 • ALTER TABLE credit_card
35232 DROP COLUMN pan;
```

Output

#	Time	Action	Message
5047	21:38:20	SELECT * FROM transactions.transaction	100001 row(s) returned
5048	21:50:30	ALTER TABLE credit_card DROP COLUMN pan	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

Query Completed

1 • `SELECT * FROM transactions.credit_card;`

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

	id	iban	pin	cvv	expiring_date
►	CcS-4857	XX4857591835292505850771	1819	467	09/27/25
	CcS-4858	XX8581768137002436094025	3964	817	12/28/28
	CcS-4859	XX7826930491423553609370	4983	277	11/26/26
	CcS-4860	XX5559590368835304645299	6876	661	07/27/27
	CcS-4861	XX2035182877195191627307	5710	398	04/25/26
	CcS-4862	XX4771731423422545422752	2842	174	11/27/25

credit_card 1 x

Exercici 8

Descarrega els arxius CSV que **trobaràs a l'apartat de recursos**:

- american_users.csv
- european_users.csv
- companies.csv
- credit_cards.csv
- transactions.csv

Estudia'ls i dissenya una base de dades amb un esquema d'estrella que contingui, almenys 4 taules de les quals puguis realitzar les següents consultes:

La taula de products.csv l'utilitzarem més endavant.

8.1 - He creado la base de datos “irinasdatabase”:

The screenshot shows a database management interface with a left sidebar, a central SQL editor, and a bottom output pane.

Left Sidebar (SCHEMAS):

- Filter objects
- gpt_filtre_agr
- gpt_join_subq
- hospitales
- irinasdatabase**
 - Tables
 - Views
 - Stored Procedures
 - Functions
- olympics
- sakila
- sys
- tienda_online
- transactions
- world
- Administration
- Schemas

Central SQL Editor:

```
1 # Nivell 1 - Exercici 8
2
3 CREATE DATABASE IF NOT EXISTS irinasdatabase;
4 USE irinasdatabase;
5
6 CREATE TABLE IF NOT EXISTS transactions (
7     id VARCHAR(60) NOT NULL,
8     card_id VARCHAR(60),
9     business_id VARCHAR(60),
10    timestamp VARCHAR(60),
11    amount DECIMAL(10,2),
12    declined TINYINT(1),
13    product_ids VARCHAR(60),
14    user_id INT,
15    lat DECIMAL(10,2),
16    longitude DECIMAL(10,2),
17    discount_amount FLOAT,
18    tax_amount FLOAT,
19    shipping_amount FLOAT,
20    channel VARCHAR(60),
21    campaign_id VARCHAR(60),
22    device_type VARCHAR(60),
```

Bottom Output Pane:

Output

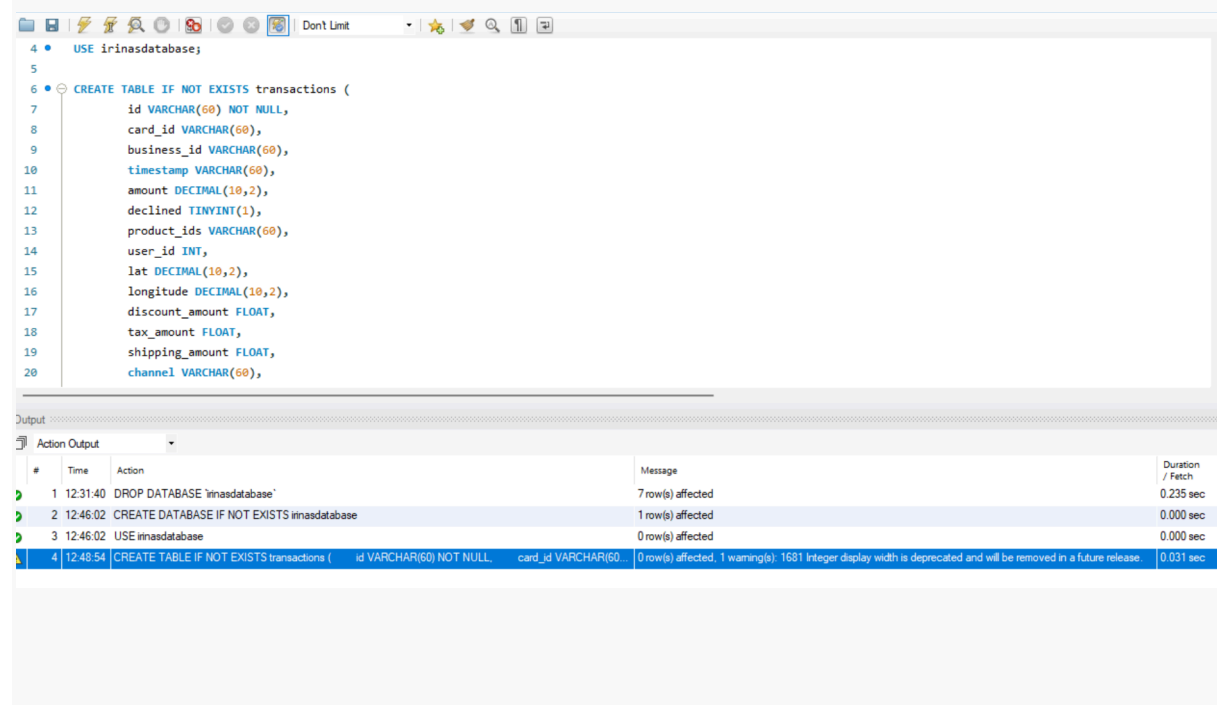
Action Output

#	Time	Action	Message	Duration / Fetch
2	12:46:02	CREATE DATABASE IF NOT EXISTS irinasdatabase	1 row(s) affected	0.000 sec

8.2 - He creado, una por una, las tablas: “transactions”, “company”, “users”, “credit_cards”. He subido datos en las tablas desde los archivos .csv proporcionados guardados previamente en C:\ProgramData\MySQL\MySQL Server 8.0

Al crear las tablas he tenido que solucionar errores relacionados con datos que superan el número máximo de caracteres previstos para un campo, con el tipo de separador del archivo csv (“,” o “;”), con el tipo de terminación de una fila (“/r/n”) con la incompatibilidad entre el tipo de datos previsto al crear la tabla y el tipo de dato (diferente) de la tabla csv a cargar.

Tabla “transactions”



The screenshot displays the MySQL Workbench interface. The top pane shows the SQL script for creating the 'transactions' table. The bottom pane shows the 'Action Output' log with four entries.

```
4 • USE irinasdatabase;
5
6 • CREATE TABLE IF NOT EXISTS transactions (
7     id VARCHAR(60) NOT NULL,
8     card_id VARCHAR(60),
9     business_id VARCHAR(60),
10    timestamp VARCHAR(60),
11    amount DECIMAL(10,2),
12    declined TINYINT(1),
13    product_ids VARCHAR(60),
14    user_id INT,
15    lat DECIMAL(10,2),
16    longitude DECIMAL(10,2),
17    discount_amount FLOAT,
18    tax_amount FLOAT,
19    shipping_amount FLOAT,
20    channel VARCHAR(60),
```

#	Time	Action	Message	Duration / Fetch
1	12:31:40	DROP DATABASE 'irinasdatabase'	7 row(s) affected	0.235 sec
2	12:46:02	CREATE DATABASE IF NOT EXISTS irinasdatabase	1 row(s) affected	0.000 sec
3	12:46:02	USE irinasdatabase	0 row(s) affected	0.000 sec
4	12:48:54	CREATE TABLE IF NOT EXISTS transactions (id VARCHAR(60) NOT NULL, card_id VARCHAR(60), business_id VARCHAR(60), timestamp VARCHAR(60), amount DECIMAL(10,2), declined TINYINT(1), product_ids VARCHAR(60), user_id INT, lat DECIMAL(10,2), longitude DECIMAL(10,2), discount_amount FLOAT, tax_amount FLOAT, shipping_amount FLOAT, channel VARCHAR(60),	0 row(s) affected, 1 warning(s): 1681 Integer display width is deprecated and will be removed in a future release	0.031 sec

```

12
13 • LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8_transactions.csv'
14 INTO TABLE transactions
15 FIELDS TERMINATED BY ','
16 ENCLOSED BY '"'
17 IGNORE 1 ROWS
18 (id, card_id, business_id, timestamp, amount, declined, product_ids, user_id, lat, longitude, discount_amount, tax_amount, shipping_amount, channel, campaign_id, device_
19
20
21 • LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8_european_users.csv'

```

#	Time	Action	Message	Duration / Fetch
24	18:10:04	TRUNCATE TABLE transactions	0 row(s) affected	0.047 sec
25	18:14:54	LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8_transactions.csv' I...	0 row(s) affected Records: 0 Deleted: 0 Skipped: 0 Warnings: 0	0.235 sec
26	18:16:15	LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8_transactions.csv' I...	0 row(s) affected Records: 0 Deleted: 0 Skipped: 0 Warnings: 0	3.922 sec
27	18:22:53	LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8_european_users.c...	0 row(s) affected Records: 0 Deleted: 0 Skipped: 0 Warnings: 0	0.031 sec
28	18:23:19	LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8_european_users.c...	3990 row(s) affected Records: 3990 Deleted: 0 Skipped: 0 Warnings: 0	0.188 sec
29	18:24:01	LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8_american_users.c...	1010 row(s) affected Records: 1010 Deleted: 0 Skipped: 0 Warnings: 0	0.047 sec

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Irina_Mihai_S2_Nocios_basi... Irina_Mihai_S4_BigQuery_Avan... users transactions credit_cards companies companies

1 • SELECT * FROM irinasdatabase.transactions;

id	card_id	business_id	timestamp	amount	declined	product_ids	user_id	lat	longitude	discount_amount	tax_amount	shipping_amount
CDDA7E40-544D-478B-A4ED-671DD8A950D9	Cc5-6894	b-2466	2018-12-12 08:05:17	176.45	0	75, 73, 98	2313	59.620510	16.559977	20.72	35.29	0
09456357-8E98-475A-8257-87A023699964	Cc5-5135	b-2342	2024-05-20 22:49:08	206.65	0	3	554	45.764588	4.843057	3.08	33.61	4.99
C47C7C84-C174-4973-A76B-825A9DB5582A	Cc5-8415	b-2250	2018-11-04 23:16:18	548.77	0	92, 85, 36, 23	3834	39.224213	-0.288066	43.76	95.24	0
2FB526AA-3844-4DDF-AC09-2EBCC25EE76	Cc5-6553	b-2610	2022-06-17 09:17:25	411.55	0	5, 27	1972	39.475985	-0.376419	4.03	71.43	0

#	Time	Action	Message	Duration / Fetch
4	20:33:25	SELECT * FROM irinasdatabase.transactions	100000 row(s) returned	0.000 sec / 0.844 sec
5	20:33:34	SELECT * FROM irinasdatabase.credit_cards	5000 row(s) returned	0.000 sec / 0.016 sec
6	20:33:42	SELECT * FROM irinasdatabase.companies	0 row(s) returned	0.000 sec / 0.000 sec
7	20:33:49	SELECT * FROM irinasdatabase.companies	0 row(s) returned	0.000 sec / 0.000 sec
8	20:34:06	LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8_companies.csv' IN...	100 row(s) affected Records: 100 Deleted: 0 Skipped: 0 Warnings: 0	0.016 sec

8:40 PM

Tabla "companies"

Filter objects

gpt_filtre_agr

gpt_join_subq

hospitales

irinasdatabase

companies

credit_card

transactions

users

Views

Stored Procedures

Functions

olympics

sakila

Administration

Schemas

Information

Table: companies

Columns:

company_id varchar(60)

company_name varchar(60)

phone varchar(60)

email varchar(60)

country varchar(60)

website varchar(60)

merchant_category varchar(60)

merchant_price_position varchar(60)

Object Info

Session

```
22 device_type VARCHAR(60),
23 is_international VARCHAR(60),
24 decline_reason VARCHAR(60),
25 distance_km DECIMAL(10,2)
26 );
27
28 CREATE TABLE IF NOT EXISTS companies (
29
30     company_id VARCHAR(60) NOT NULL,
31     company_name VARCHAR(60),
32     phone VARCHAR(60),
33     email VARCHAR(60),
34     country VARCHAR(60),
35     website VARCHAR(60),
36     merchant_category VARCHAR(60),
37     merchant_price_position VARCHAR(60)
38 );
39
```

Output

#	Time	Action	Message	Duration / Fetch
1	19:40:56	ALTER TABLE companies RENAME COLUMN id TO company_id	Error Code: 1046. No database selected Select the default DB to be used by doubl...	0.000 sec
2	19:41:04	ALTER TABLE companies RENAME COLUMN id TO company_id	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
3	19:41:55	LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex...	Error Code: 1054. Unknown column 'merchant_category' in field list'	0.000 sec
4	19:43:00	DROP TABLE companies	0 row(s) affected	0.016 sec
5	19:46:02	CREATE TABLE IF NOT EXISTS companies (company_id VARCHAR(60)...	0 row(s) affected	0.016 sec

irinasdatabase

companies

credit_cards

transactions

users

Columns

Indexes

Foreign Keys

Triggers

Views

Stored Procedures

Functions

olympics

Administration

Schemas

Information

Table: companies

Columns:

company_id varchar(60)

company_name varchar(60)

phone varchar(60)

email varchar(60)

country varchar(60)

website varchar(60)

merchant_category varchar(60)

merchant_price_position varchar(60)

Object Info

Session

```
75 FIELDS TERMINATED BY ',';
76 ENCLOSED BY '"'
77 IGNORE 1 ROWS
78 (id, card_id, business_id, timestamp, amount, declined, product_ids, user_id, lat, longitude, discount_amount, tax_amount, shipping
79
80 LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex-8_companies.csv'
81 INTO TABLE companies
82 FIELDS TERMINATED BY ','
83 ENCLOSED BY '"'
84 IGNORE 1 ROWS
85 (company_id, company_name, phone, email, country, website, merchant_category, merchant_price_position);
86
87
88 LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex-8_european_users.csv'
89 INTO TABLE users
90 FIELDS TERMINATED BY ','
91 ENCLOSED BY '"'
```

Output

#	Time	Action	Message	Duration / Fetch
4	20:33:25	SELECT * FROM irinasdatabase.transactions	100000 row(s) returned	0.000 sec / 0.844
5	20:33:34	SELECT * FROM irinasdatabase.credit_cards	5000 row(s) returned	0.000 sec / 0.016
6	20:33:42	SELECT * FROM irinasdatabase.companies	0 row(s) returned	0.000 sec / 0.000
7	20:33:49	SELECT * FROM irinasdatabase.companies	0 row(s) returned	0.000 sec / 0.000
8	20:34:06	LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex...	100 row(s) affected Records: 100 Deleted: 0 Skipped: 0 Warnings: 0	0.016 sec

1

SELECT * FROM irinasdatabase.companies;

Result Grid

Filter Rows:

Export:

Wrap Cell Contents:

	company_id	company_name	phone	email	country	website	merchant_category	merchant_price_position
b-2222		Ac Fermentum Incorporated	+64 601 244 6320	donec.porttitor.tellus@yahoo.net	Germany	https://www.ac-fermentum-incorporated.com	Beauty	mid-market
b-2226		Magna A Neque Industries	+33 823 560 3779	risus.donec.nibh@icloud.org	Australia	https://www.magna-a-neque-industries.com	Travel	budget
b-2230		Fusce Corp.	+29 657 646 4324	risus@protonmail.edu	United States	https://www.fusce-corp.com	Home	mid-market
b-2234		Convallis In Incorporated	+6 693 982 1255	mauris.ut@aol.co.uk	Germany	https://www.convallis-in-incorporated.com	Sports	mid-market
b-2238		Ante Iaculis Nec Foundation	+69 914 549 1009	sed.dictum.proin@outlook.ca	New Zealand	https://www.ante-iaculis-nec-foundation.com	Toys	mid-market

companies 1 x

Tabla "credit_cards"

SCHEMAS

Filter objects

- irinasdatabase
 - Tables
 - companies
 - credit_cards**
 - transactions
 - users
 - Views
 - Stored Procedures
 - Functions
- olympics
- sakila
- sys
- tienda_online
- transactions

Administration Schemas

Information

Table: credit_cards

Columns:

id	varchar(60)
user_id	int
iban	varchar(60)
pan	varchar(60)
pin	int
cvv	int
track1	varchar(255)
track2	varchar(255)
expiring_date	varchar(60)
card_type	varchar(60)
card_renewal_flag	tinyint

```
56
57
58 CREATE TABLE IF NOT EXISTS credit_cards (
59     id VARCHAR(60) NOT NULL,
60     user_id INT NOT NULL,
61     iban VARCHAR(60),
62     pan VARCHAR(60),
63     pin INT,
64     cvv INT,
65     track1 VARCHAR(255),
66     track2 VARCHAR(255),
67     expiring_date VARCHAR(60),
68     card_type VARCHAR(60),
69     card_renewal_flag TINYINT
70 );
71
72
73 LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__transactions.csv'
```

Output

#	Time	Action	Message	Duration / Fetch
1	20:25:18	CREATE TABLE IF NOT EXISTS credit_cards (id VARCHAR(60) NOT NULL...	0 row(s) affected	0.031 sec
2	20:25:35	LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex...	5000 row(s) affected Records: 5000 Deleted: 0 Skipped: 0 Warnings: 0	0.156 sec

SCHEMAS

Filter objects

- irinasdatabase
 - Tables
 - companies
 - credit_cards**
 - transactions
 - users
 - Views
 - Stored Procedures
 - Functions
- olympics
- sakila
- sys
- tienda_online
- transactions

Administration Schemas

Information

Table: credit_cards

Columns:

id	varchar(60)
user_id	int
iban	varchar(60)
pan	varchar(60)
pin	int
cvv	int
track1	varchar(255)
track2	varchar(255)
expiring_date	varchar(60)
card_type	varchar(60)
card_renewal_flag	tinyint

```
95
96 LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__american_users.csv'
97 INTO TABLE users
98 FIELDS TERMINATED BY ','
99 ENCLOSED BY '"'
100 IGNORE 1 ROWS
101 (id,name,surname,phone,email,birth_date,country,city,postal_code,address,signup_date,user_segment,income_band)
102 SET region = "America";
103
104 LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__credit_cards.csv'
105 INTO TABLE credit_cards
106 FIELDS TERMINATED BY ','
107 ENCLOSED BY '"'
108 IGNORE 1 ROWS
109 (id,user_id,iban,pan,pin,cvv,track1,track2,expiring_date,card_type,card_renewal_flag)
110
111 SHOW CREATE TABLE credit_cards
```

Output

#	Time	Action	Message	Duration / Fetch
1	20:25:18	CREATE TABLE IF NOT EXISTS credit_cards (id VARCHAR(60) NOT NULL...	0 row(s) affected	0.031 sec
2	20:25:35	LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex...	5000 row(s) affected Records: 5000 Deleted: 0 Skipped: 0 Warnings: 0	0.156 sec

1 • SELECT * FROM irinasdatabase.credit_cards;

Result Grid												Result Grid
Filter Rows: Exports: Wrap Cell Contents: Fetch rows:												
	id	user_id	iban	pan	pin	cvv	track1	track2	expiring_date	card_type	card_u	
▶	CcU-2938	275	CA13755052951024363769804	5424465566813633	3257	984	%88383712448554646^WovsxejDpwiev^8604...	%87653863056044187=800716333673	04/27/28	credit	1	
	CcU-2945	274	CA657195557325315948524978	5142423821948828	9080	887	%84621311609958661^UftuyfsSeimwn^06106...	%84149568437843501=510714033071	01/27/28	premium_credit	1	
	CcU-2952	273	CA749863982219203741201012	4556453555387	4598	438	%82183285104307501^CddyrtUxwfdq^5907...	%86778580257827162=6906859740077	08/29/28	credit	1	
	CcU-2959	272	CA164763652898476863375896	372461377349375	3583	667	%87281111956795320^XocddjBkecd^09016...	%84246154489281853=280522391678	08/29/28	credit	1	

Tabla “users”

Los archivos “european_users” y “american_users” los he juntado en un solo archivo “users”, ya que se trata de ficheros con estructura idéntica. Para poder filtrar por continente, he creado una columna “region” en la tabla “users”.

```
36
37 CREATE TABLE IF NOT EXISTS users (
38     id VARCHAR(60),
39     name VARCHAR(60),
40     surname VARCHAR(60),
41     phone VARCHAR(60),
42     email VARCHAR(60),
43     birth_date VARCHAR(60),
44     country VARCHAR(60),
45     city VARCHAR(60),
46     postal_code VARCHAR(60),
47     address VARCHAR(60),
48     signup_date DATE,
49     user_segment VARCHAR(60),
50     income_band VARCHAR(10),
51     region VARCHAR(10)
52 );
```

Output				
Action Output				
#	Time	Action	Message	Duration / Fetch
3	12:46:02	USE itinasdatabase	0 row(s) affected	0.000 sec
4	12:48:54	CREATE TABLE IF NOT EXISTS transactions (id VARCHAR(60) NOT NULL, card_id VARCHAR(60) NOT NULL, company_name VARCHAR(60) NOT NULL, transaction_date DATE, transaction_amount DECIMAL(10,2), transaction_type VARCHAR(10), transaction_status VARCHAR(10), transaction_notes VARCHAR(255))	0 row(s) affected, 1 warning(s): 1681 Integer display width is deprecated and will be removed in a future release.	0.031 sec
5	12:53:59	CREATE TABLE IF NOT EXISTS company (id VARCHAR(15) NOT NULL, company_name VARCHAR(60), company_address VARCHAR(255), company_phone VARCHAR(20), company_email VARCHAR(60), company_website VARCHAR(60), company_status VARCHAR(10), company_notes VARCHAR(255))	0 row(s) affected	0.016 sec
6	12:54:25	CREATE TABLE IF NOT EXISTS users (id VARCHAR(60), name VARCHAR(60), surname VARCHAR(60), phone VARCHAR(20), email VARCHAR(60), birth_date DATE, country VARCHAR(60), city VARCHAR(60), postal_code VARCHAR(10), address VARCHAR(255), signup_date DATE, user_segment VARCHAR(60), income_band VARCHAR(10), region VARCHAR(10))	0 row(s) affected	0.031 sec
7	12:54:33	CREATE TABLE IF NOT EXISTS credit_card (id VARCHAR(8) PRIMARY KEY, iban VARCHAR(20), cardholder_name VARCHAR(60), cardholder_surname VARCHAR(60), cardholder_phone VARCHAR(20), cardholder_email VARCHAR(60), cardholder_address VARCHAR(255), cardholder_signup_date DATE, cardholder_user_segment VARCHAR(60), cardholder_income_band VARCHAR(10), cardholder_region VARCHAR(10))	0 row(s) affected, 2 warning(s): 1681 Integer display width is deprecated and will be removed in a future release.	0.015 sec

```

LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__european_users.csv'
INTO TABLE users
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
IGNORE 1 ROWS
(id,name,surname,phone,email,birth_date,country,city,postal_code,address,signup_date,user_segment,income_band)
SET region = "Europe";

LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__american_users.csv'
INTO TABLE users
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
IGNORE 1 ROWS
(id,name,surname,phone,email,birth_date,country,city,postal_code,address,signup_date,user_segment,income_band)
SET region = "America";

```

Irina_Mihai_S2_Nocios_basiqu... Irina_Mihai_S4_BigQuery_Avan... users transactions credit_cards companies

Don't Limit

1 • SELECT * FROM irinasdatabase.users;

Result Grid

Filter Rows:

Export:

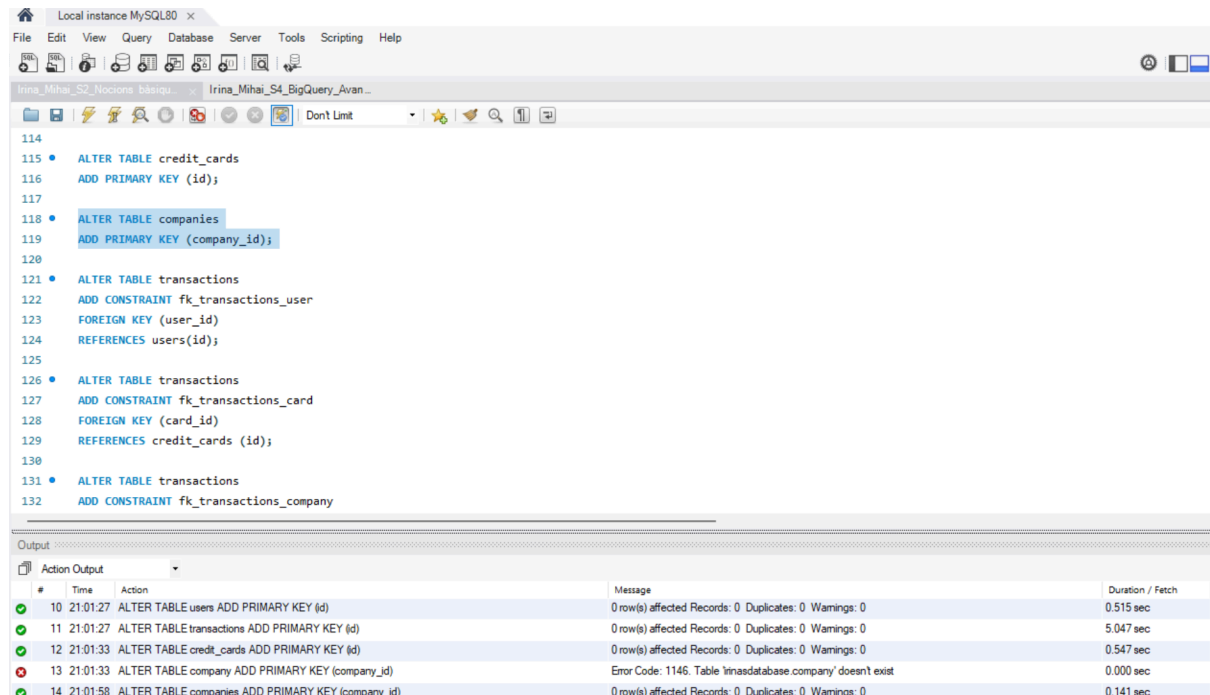
Wrap Cell Contents:

Fetch rows:

	surname	phone	email	birth_date	country	city	postal_code	address	signup_date	user_segment	income_band	region
	Uugjkvne	+97-556-4320	xmjnxt.uugjkvne@example.com	Feb 27, 2002	France	Marseille	13001	748 Uugjkvne St	2014-01-25	student	low	Europe
	Vpkviadx	+67-763-3511	yxkfvb.vpkviadx@example.com	Sep 26, 1957	Spain	Sevilla	41001	886 Vpkviadx St	2014-01-29	established	high	Europe
	Gamble	1-282-581-0551	interdum.enim@protonmail.edu	Nov 17, 1985	United States	New York	10001	348-7818 Sagittis St.	2014-04-23	young_professional	high	America
	Mcconnell	(718) 257-2412	integer.vitae.nibh@protonmail.org	Aug 23, 1992	United States	Philadelphia	19101	903 Sit Ave	2014-07-04	young_professional	high	America

users 1 x

8.3 - He añadido las primary keys y he creado las relaciones entre las tablas (CONSTRAINTS). Al intentar crear la relación entre “transactions” y “users” me he encontrado que los campos de las foreign keys no eran compatibles, así que he modificado el tipo de datos de “transactions.user_id” de INT a VARCHAR(60).



The screenshot shows the MySQL Workbench interface with a SQL editor and an output window. The SQL editor contains the following queries:

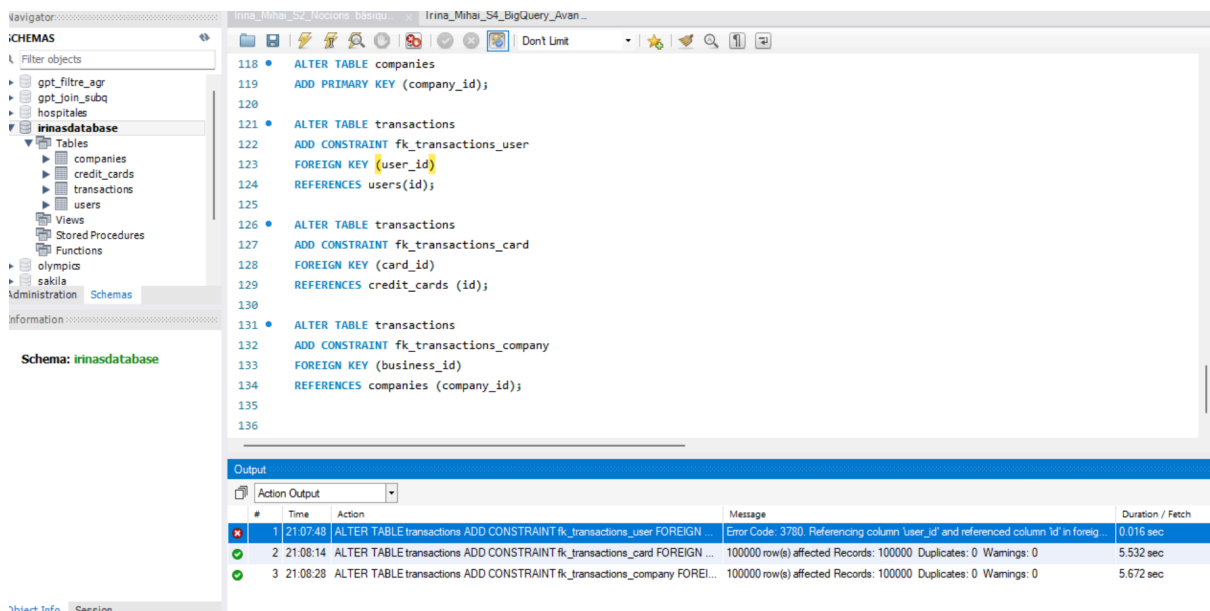
```

114
115 • ALTER TABLE credit_cards
116   ADD PRIMARY KEY (id);
117
118 • ALTER TABLE companies
119   ADD PRIMARY KEY (company_id);
120
121 • ALTER TABLE transactions
122   ADD CONSTRAINT fk_transactions_user
123   FOREIGN KEY (user_id)
124   REFERENCES users(id);
125
126 • ALTER TABLE transactions
127   ADD CONSTRAINT fk_transactions_card
128   FOREIGN KEY (card_id)
129   REFERENCES credit_cards (id);
130
131 • ALTER TABLE transactions
132   ADD CONSTRAINT fk_transactions_company

```

The output window shows the results of these queries:

#	Time	Action	Message	Duration / Fetch
10	21:01:27	ALTER TABLE users ADD PRIMARY KEY (id)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.515 sec
11	21:01:27	ALTER TABLE transactions ADD PRIMARY KEY (id)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	5.047 sec
12	21:01:33	ALTER TABLE credit_cards ADD PRIMARY KEY (id)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.547 sec
13	21:01:33	ALTER TABLE company ADD PRIMARY KEY (company_id)	Error Code: 1146. Table 'ininasdatabase.company' doesn't exist	0.000 sec
14	21:01:58	ALTER TABLE companies ADD PRIMARY KEY (company_id)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.141 sec



The screenshot shows the MySQL Workbench interface with the database schema visible on the left and SQL queries in the editor. The schema shows the 'ininasdatabase' with tables: companies, credit_cards, transactions, and users. The SQL editor contains the following queries:

```

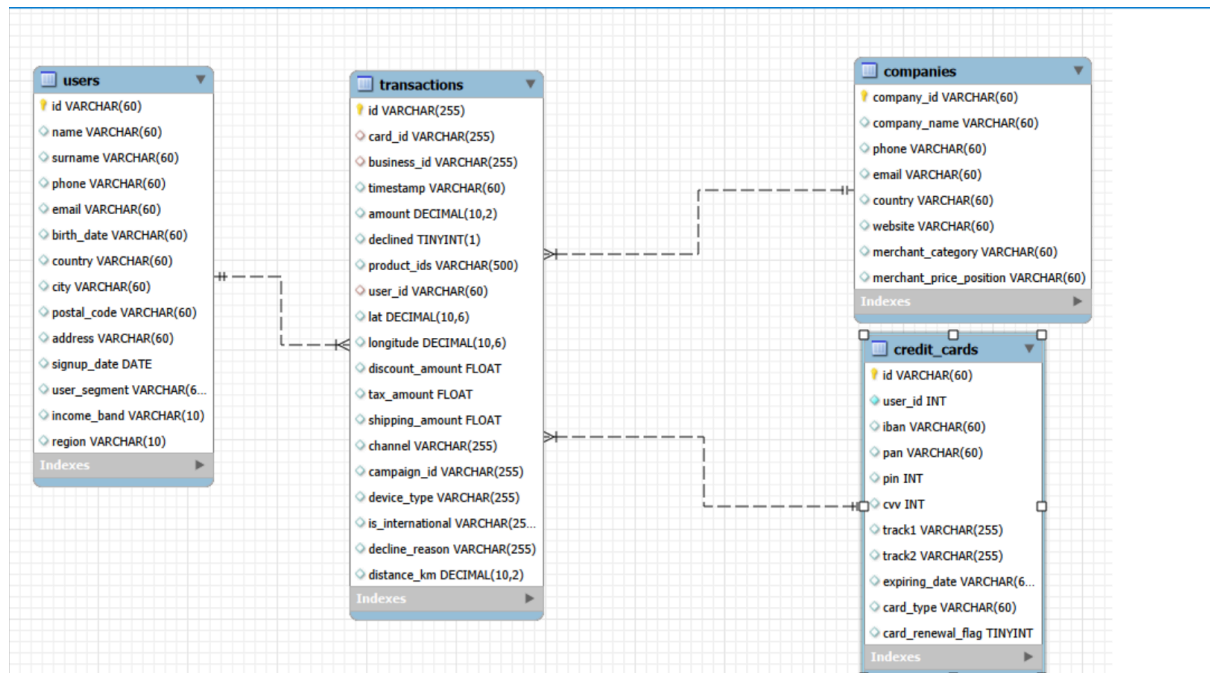
118 • ALTER TABLE companies
119   ADD PRIMARY KEY (company_id);
120
121 • ALTER TABLE transactions
122   ADD CONSTRAINT fk_transactions_user
123   FOREIGN KEY (user_id)
124   REFERENCES users(id);
125
126 • ALTER TABLE transactions
127   ADD CONSTRAINT fk_transactions_card
128   FOREIGN KEY (card_id)
129   REFERENCES credit_cards (id);
130
131 • ALTER TABLE transactions
132   ADD CONSTRAINT fk_transactions_company
133   FOREIGN KEY (business_id)
134   REFERENCES companies (company_id);
135
136

```

The output window shows the results of these queries:

#	Time	Action	Message	Duration / Fetch
1	21:07:48	ALTER TABLE transactions ADD CONSTRAINT fk_transactions_user FOREIGN...	Error Code: 3780. Referencing column 'user_id' and referenced column 'id' in foreig...	0.016 sec
2	21:08:14	ALTER TABLE transactions ADD CONSTRAINT fk_transactions_card FOREIGN ...	100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0	5.532 sec
3	21:08:28	ALTER TABLE transactions ADD CONSTRAINT fk_transactions_company FOREI...	100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0	5.672 sec

En el diagrama de abajo se ven las relaciones de 1 a N que he creado entre las 3 tablas de dimensiones “users”, “companies” y “credit_cards” y la tabla de hechos “transactions”. Las llaves amarillas dentro de cada tabla representan las primary keys de cada tabla. Los rombos rojos dentro de “transactions” representan las claves foraneas que relacionan la tabla con las otras 3 tablas.



Exercici 9

Realitza una subconsulta que mostri tots els usuaris amb més de 80 transaccions utilitzant almenys 2 taules. (consulta optimizada según recomendación)

```
221
222
223 # Nivell 1 - Exercici 9
224
225 # Realitza una subconsulta que mostri tots els usuaris amb més de 80 transaccions utilitzant almenys 2 taules.
226
227 • SELECT id, name, surname
228 FROM users as u
229 WHERE id IN (
230     SELECT user_id FROM transactions AS t GROUP BY t.user_id HAVING COUNT(id) > 80);
231
```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

id	name	surname
185	Molly	Gilliam
289	Dxwgi	Hwcru
318	Bnyr	Astuw
454	Sfzzoh	Xgvfridxs
NULL	NULL	NULL

users 21

Result Grid

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 14	15:01:08	SHOW CREATE TABLE companies	1 row(s) returned	0.031 sec / 0.000 sec
✓ 15	15:04:02	SHOW CREATE TABLE credit_cards	1 row(s) returned	0.016 sec / 0.000 sec
✓ 16	15:12:41	SELECT id, name, surname FROM users as u WHERE id IN (SELECT user_id ...	4 row(s) returned	1.000 sec / 0.000 sec

Exercici 10

Mostra la mitjana d'amount per IBAN de les targetes de crèdit a la companyia Donec Ltd, utilitza almenys 2 taules.

The screenshot shows a SQL IDE interface with a query editor and a results pane. The query editor contains the following SQL code:

```
55333 # Mostra la mitjana d'amount per IBAN de les targetes de crèdit a la companyia Donec Ltd, utilitza almenys 2 taules.
55334
55335 • SELECT AVG(amount), iban
55336 FROM transactions
55337 JOIN credit_cards
55338 ON credit_cards.id = transactions.card_id
55339 JOIN companies
55340 ON companies.company_id = transactions.business_id
55341 WHERE company_name = "Donec Ltd"
55342 GROUP BY iban;
```

The results pane displays a table with two columns: **AVG(amount)** and **iban**. The table contains 10 rows of data. Below the table, the 'Output' pane shows the execution log with two entries:

#	Time	Action	Message
33	12:41:24	SELECT AVG(amount), iban FROM transactions JOIN credit_cards ON credit_cards.id = transactions.car...	5000 row(s) returned
34	12:42:55	SELECT AVG(amount), iban FROM transactions JOIN credit_cards ON credit_cards.id = transactions.car...	371 row(s) returned

Query Completed

Nivell 2 - Exercici 1

Identifica els cinc dies que es va generar la quantitat més gran d'ingressos a l'empresa per vendes. Mostra la data de cada transacció juntament amb el total de les vendes.

He añadido WHERE declined=0

```
251 # Identifica els cinc dies que es va generar la quantitat més gran d'ingressos a l'empresa per vendes.
252 # Mostra la data de cada transacció juntament amb el total de les vendes.
253
254 • SELECT DATE(timestamp) as Dia, SUM(amount) AS Total_vendes_per_dia
255 FROM transactions
256 WHERE declined=0
257 GROUP BY DATE(timestamp)
258 ORDER BY Total_vendes_per_dia desc
259 LIMIT 5
260
261 # Nivell 2 - Exercici 2
```

Result Grid | Filter Rows: | Exports: | Wrap Cell Content: |

	Dia	Total_vendes_per_dia
▶	2023-02-20	15728.399993896484
	2019-11-18	15447.8900680542
	2022-12-13	14944.000034332275
	2019-03-18	14896.47006225586
	2017-12-20	14367.95998764038

Result 22 x

Output

Action Output

#	Time	Action	Message	Duration
✓ 15	15:04:02	SHOW CREATE TABLE credit_cards	1 row(s) returned	0.016 sec
✓ 16	15:12:41	SELECT id, name, surname FROM users as u WHERE id IN (SELECT user_id ...	4 row(s) returned	1.000 sec
✓ 17	15:18:50	SELECT DATE(timestamp) as Dia, SUM(amount) AS Total_vendes_per_dia FR...	5 row(s) returned	0.391 sec

Exercici 2

Presenta el nom, telèfon, país, data i amount, d'aquelles empreses que van realitzar transaccions amb un valor comprès entre 350 i 400 euros i en alguna d'aquestes dates: 29 d'abril del 2015, 20 de juliol del 2018 i 13 de març del 2024. Ordena els resultats de major a menor quantitat.

Irina_Mihai_Lasca SZUT. Nocio... x companies transactions credit_cards transaction_products

5379 # d'aquelles empreses que van realitzar transaccions amb un valor comprès entre 350 i 400 euros
 5380 # i en alguna d'aquestes dates: 29 d'abril del 2015, 20 de juliol del 2018 i 13 de març del 2024.
 5381 # Ordena els resultats de major a menor quantitat.
 5382
 5383 SELECT company_name, phone, country, amount
 5384 FROM companies
 5385 JOIN transactions
 5386 ON companies.company_id = transactions.business_id
 5387 WHERE timestamp IN (SELECT DATE(timestamp) FROM transactions WHERE DATE(timestamp) IN ("2015-04-29", "2018-07-20", "2024-03-13") AND
 5388 AND amount BETWEEN 350 and 400
 5389 ORDER BY amount desc;
 5390
 5391
 5392
 5393 # Exercici 3

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

company_name	phone	country	amount
Fringilla LLC	+1 150 784 1716	New Zealand	390.33
Auctor Mauris Vel LLP	+32 133 447 8239	United States	383.58
Mauris Institute	+75 473 619 9472	Sweden	369.2
Pede Cum Ltd	+26 625 798 6126	Norway	358.64
Netus Et Malesuada Ltd	+52 818 402 1638	Netherlands	352.21

Result 5 x

Output

Action Output

#	Time	Action	Message
7	21:39:10	SELECT company_name, phone, country, amount FROM companies JOIN transactions ON companies.c...	6 row(s) returned
8	21:39:47	SELECT company_name, phone, country, amount FROM companies JOIN transactions ON companies.c...	6 row(s) returned

Exercici 3

Necessitem optimitzar l'assignació dels recursos i dependrà de la capacitat operativa que es requereixi, per la qual cosa et demanen la informació sobre la quantitat de transaccions que realitzen les empreses, però el departament de recursos humans és exigent i vol un llistat de les empreses on especifiquis si tenen igual o més de 400 transaccions o menys.

```

371 # Necessitem optimitzar l'assignació dels recursos i dependrà de la capacitat operativa que es requereixi,
372 # per la qual cosa et demanen la informació sobre la quantitat de transaccions que realitzen les empreses,
373 # però el departament de recursos humans és exigent i vol un llistat de les empreses
374 # on especifiquis si tenen igual o més de 400 transaccions o menys.
375 • SELECT companies.company_name, t1.purchases,
376 CASE
377     WHEN t1.purchases > 400 THEN "Above 400"
378     WHEN t1.purchases < 400 THEN "Below 400"
379     WHEN t1.purchases = 400 THEN "400"
380 END AS transaction_level
381 FROM (SELECT COUNT(id) AS purchases, business_id FROM transactions GROUP BY business_id) as t1
382 JOIN companies
383 ON companies.company_id = t1.business_id;
384

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

company_name	purchases	transaction_level
Sed Nunc Ltd	1541	Above 400
Amet Nulla Donec Corporation	1511	Above 400
Nascetur Ridiculus Mus Inc.	424	Above 400
Vestibulum Lorem PC	450	Above 400
Gravida Sagittis LLP	1475	Above 400

Result 22 x

Output

Action Output

#	Time	Action	Message
67	13:42:19	SELECT companies.company_name,t1.purchases CASE WHEN t1.purchases > 400 THEN "Above 400..."	Error Code: 1064. You have an error in your SQL syntax; check the...
68	13:43:21	SELECT companies.companv name,t1.purchases CASE WHEN t1.purchases > 400 THEN "Above 40..."	100 row(s) returned

Completed

Exercici 4

Elimina de la taula transaction el registre amb ID

000447FE-B650-4DCF-85DE-C7ED0EE1CAAD de la base de dades.

```

:383 ON companies.company_id = t1.business_id;
:384
:385
:386
:387
:388
:389
:390
:391
:392 # Exercici 4
:393
:394 # Elimina de la taula transaction el registre amb ID 000447FE-B650-4DCF-85DE-C7ED0EE1CAAD de la base de dades.
:395
:396 • DELETE FROM transactions
:397 WHERE id = "000447FE-B650-4DCF-85DE-C7ED0EE1CAAD"
:398
:399
:400
:401
:402
:403

```

Output

#	Time	Action	Message
68	13:43:21	SELECT companies.company_name,t1.purchases, CASE WHEN t1.purchases > 400 THEN "Above 400"	100 row(s) returned
69	13:53:12	DELETE FROM transactions WHERE id = "000447FE-B650-4DCF-85DE-C7ED0EE1CAAD"	1 row(s) affected

Query Completed

Exercici 5 -he añadido el código “Create or replace view VM AS”

La secció de màrqueting desitja tenir accés a informació específica per a realitzar anàlisi i estratègies efectives. S'ha sol·licitat crear una vista que proporcioni detalls clau sobre les companyies i les seves transaccions.

Serà necessària que creïs una vista anomenada VistaMarketing que contingui la següent informació: Nom de la companyia. Telèfon de contacte. País de residència. Mitjana de compra realitzat per cada companyia. Presenta la vista creada, ordenant les dades de major a menor mitjana de compra.

```
File Edit View Query Database Server Tools Scripting Help
Irina_Mihai_S3_Introductio_Big... Irina_Mihai_S2_Nociosn_bàsiqu... transaction credit_card company transaction irinasdatabase.vistamarketing
Don't Limit
342 # Serà necessària que creis una vista anomenada VistaMarketing
343 # que contingui la següent informació:
344 # Nom de la companyia. Telèfon de contacte. País de residència. Mitjana de compra realitzat per cada companyia.
345 # Presenta la vista creada, ordenant les dades de major a menor mitjana de compra.
346
347 CREATE OR REPLACE VIEW VistaMarketing AS
348 SELECT company_name, phone, country, AVG(amount) AS mitjana_compra
349 FROM companies
350 JOIN transactions
351 ON companies.company_id = transactions.business_id
352 GROUP BY company_name, phone, country
353 ORDER BY mitjana_compra desc;
354
355
356 # Nivell 3 - Exercici 1
357
358 # Crea una nova taula que reflecteixi l'estat de les targetes de crèdit
359 # basat en si les tres últimes transaccions han estat declinades aleshores és inactiu,
360 # si almenys una no és rebutjada aleshores és actiu.
```

Output				
Action Output				
#	Time	Action	Message	Duration / Fetch
16	15:12:41	SELECT id, name, surname FROM users as u WHERE id IN (SELECT user_id FROM transactions AS t ...	4 row(s) returned	1.000 sec / 0.000 sec
17	15:18:50	SELECT DATE(timestamp) as Dia, SUM(amount) AS Total_vendes_per_dia FROM transactions WHERE ...	5 row(s) returned	0.391 sec / 0.000 sec
18	15:46:41	CREATE OR REPLACE VIEW VistaMarketing AS SELECT company_name, phone, country, AVG(amoun...	0 row(s) affected	0.047 sec

Result Grid			
Filter Rows:	Export:	Wrap Cell Content:	
company_name	phone	country	mitjana_compra
Ac Fermentum Incorporated	+64 601 244 6320	Germany	330.7693381388949
Ut Semper Foundation	+10 283 500 6449	Sweden	307.3729452766784
Orci Adipiscing Limited	+60 897 390 5820	United Kingdom	306.2656474370597
Dolor Vitae Limited	+6 421 567 9195	France	305.9527446211815
Nunc In Foundation	+13 267 741 2619	Italy	305.3961408791928

Result 6 x

Output				
Action Output				
#	Time	Action	Message	
8	21:39:47	SELECT company_name, phone, country, amount FROM companies JOIN transactions ON companies.c...	6 row(s) returned	
9	21:41:48	SELECT company_name, phone, country, AVG(amount) AS mitjana_compra FROM companies JOIN tran...	100 row(s) returned	

Navigator: Irina_Mihai_Tasca S2.01. Nocio... companies transactions credit_cards vistamarketing - View x vistamarketing

SCHEMAS

Filter objects

- gpt_filtre_agr
- gpt_join_subq
- hospitales
- irinasdatabase
 - Tables
 - companies
 - credit_cards
 - transactions
 - users
 - Views
 - vistamarketing
 - Stored Procedures
 - Functions
 - olympics
 - sakila

Administration Schemas

Information

View: vistamarketing

Columns:

company_name	varchar(60)
phone	varchar(60)
country	varchar(60)
mitjana_compra	double

Name: vistamarketing

The name of the view is parsed automatically from the DDL statement. The DDL is parsed automatically while you type.

DDL:

```

1 • CREATE
2     ALGORITHM = UNDEFINED
3     DEFINER = `root`@`localhost`
4     SQL SECURITY DEFINER
5     VIEW `vistamarketing` AS
6     SELECT
7         `companies`.`company_name` AS `company_name`,
8         `companies`.`phone` AS `phone`,
9         `companies`.`country` AS `country`,
10        AVG(`transactions`.`amount`) AS `mitjana_compra`
11    FROM
12        (`companies`
13         JOIN `transactions` ON ((`companies`.`company_id` = `transactions`.`business_id`)))
14    GROUP BY `companies`.`company_name`, `companies`.`phone`, `companies`.`country`
15    ORDER BY `mitjana_compra` DESC
  
```

View

Output

Action Output

#	Time	Action	Message	Duration / Fetch
75	19:19:37	Apply changes to vistamarketing	Changes applied	
76	19:20:06	SELECT * FROM irinasdatabase.vistamarketing	100 row(s) returned	0.828 sec / 0.000 sec

Object Info Session

uerv Completed

Navigator: Irina_Mihai_Tasca S2.01. Nocio... companies transactions credit_cards vistamarketing - View x vistamarketing

SCHEMAS

Filter objects

- gpt_filtre_agr
- gpt_join_subq
- hospitales
- irinasdatabase
 - Tables
 - companies
 - credit_cards
 - transactions
 - users
 - Views
 - vistamarketing
 - Stored Procedures
 - Functions
 - olympics
 - sakila

Administration Schemas

Information

View: vistamarketing

Columns:

company_name	varchar(60)
phone	varchar(60)
country	varchar(60)
mitjana_compra	double

1 • SELECT * FROM irinasdatabase.vistamarketing;

Result Grid

company_name	phone	country	mitjana_compra
Ac Fermentum Incorporated	+64 601 244 6320	Germany	330.7693381388949
Ut Semper Foundation	+10 283 500 6449	Sweden	307.3729452766784
Orci Adipiscing Limited	+60 897 390 5820	United Kingdom	306.2656474370597
Dolor Vitae Limited	+6 421 567 9195	France	305.9527446211815
Nunc In Foundation	+13 267 741 2619	Italy	305.3961408791928
Maecenas Malesuada Fringilla Inc.	+51 673 540 5289	Netherlands	305.1497312375065
Aliquet Sem Limited	+44 744 425 9915	Netherlands	304.2117177984391
Nulla Integer Vulputate Corp.	+75 236 687 9162	Sweden	303.4248790811719
Non Justo Corp.	+67 963 294 1666	Sweden	303.3045736373842

stamarketing 1 x

Output

Action Output

#	Time	Action	Message
75	19:19:37	Apply changes to vistamarketing	Changes applied
76	19:20:06	SELECT * FROM irinasdatabase.vistamarketing	100 row(s) returned

Object Info Session

Nivell 3 - Exercici 1

Crea una nova taula que reflecteixi l'estat de les targetes de crèdit basat en si les tres últimes transaccions han estat declinades aleshores és inactiu, si almenys una no és rebutjada aleshores és actiu. Partint d'aquesta taula respon: Quantes targetes estan actives?

```
Irina_Mihai_Tasca S2.01. Nocio... companies transactions credit_cards
SELECT COUNT(card_id),
CASE
  WHEN SUM(declined=0) = 0 THEN "Inactive"
  ELSE "Active"
END AS situacio
FROM (
  SELECT card_id, timestamp, declined,
  ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY timestamp DESC) AS R_N
  FROM transactions) AS posicio
WHERE R_N <= 3;
```

Result Grid

COUNT(card_id)	situacio
15000	Active

Result 34 x

Output

#	Time	Action	Message
92	21:54:45	SELECT COUNT(card_id), CASE WHEN SUM(declined=0) = 0 THEN "Inactive" ELSE "Active" END AS...	5000 row(s) returned
93	21:55:13	SELECT COUNT(card_id), CASE WHEN SUM(declined=0) = 0 THEN "Inactive" ELSE "Active" END AS...	1 row(s) returned

Query Completed

Exercici 2

Crea una taula amb la qual puguem unir les dades de l'arxiu de products.csv amb la base de dades creada (ja que fins ara no podem fer-ho), tenint en compte que des de transaction tens product_ids. Genera la següent consulta: Necessitem conèixer el nombre de vegades que s'ha venut cada producte.

2.1 - He creado la nueva tabla.

The screenshot displays the SQL Server Enterprise Manager interface. The left pane shows the 'irinasdatabase' schema with tables like 'companies', 'credit_cards', 'products', 'transactions', and 'users'. The central pane contains a SQL script for creating a 'products' table. The right pane shows the output of the script execution.

```
35431 # Exercici 2
35432 # Crea una taula amb la qual puguem unir les dades de l'arxiu de products.csv
35433 # amb la base de dades creada
35434 # (ja que fins ara no podiem fer-ho),
35435 # tenint en compte que des de transaction tens product_ids.
35436 # Genera la següent consulta:
35437 # Necessitem conèixer el nombre de vegades que s'ha venut cada producte.
35438
35439 CREATE TABLE IF NOT EXISTS products (
35440     id INT,
35441     product_name VARCHAR(60),
35442     price DECIMAL(10, 2),
35443     colour VARCHAR(60),
35444     weight FLOAT,
35445     warehouse_id VARCHAR(60),
35446     category VARCHAR(60),
35447     brand VARCHAR(60),
35448     cost DECIMAL(10, 2),
35449     launch_date DATE
35450 );
35451
```

Output

#	Time	Action	Message	Duration / Fetch
96	23:34:45	DROP TABLE 'irinasdatabase'.products'	0 row(s) affected	0.015 sec
97	23:36:04	CREATE TABLE IF NOT EXISTS products (id INT, product name VARCHA...	0 row(s) affected	0.000 sec

2.2 - He importado los datos teniendo en cuenta el carácter “\$”

The screenshot shows the MySQL Workbench interface. On the left, the 'Schemas' pane shows the 'irinasdatabase' with a 'products' table. Below it, the 'Table: products' structure is listed with columns: id (int), product_name (varchar(60)), price (decimal(10,2)), colour (varchar(60)), weight (float), warehouse_id (varchar(60)), category (varchar(60)), brand (varchar(60)), cost (decimal(10,2)), and launch_date (date).

The main editor shows a SQL query (Query 1) with the following lines:

```
30 FIELDS TERMINATED BY ','
31 ENCLOSED BY '"'
32 LINES TERMINATED BY '\n'
33 IGNORE 1 ROWS;
34
35 LOAD DATA LOCAL INFILE 'C:/Users/irina/Desktop/DATA/DATA ANALIST IT ACADEMY/ESPECIALITZACIO...'
36 INTO TABLE products
37 FIELDS TERMINATED BY ','
38 OPTIONALLY ENCLOSED BY '"'
39 LINES TERMINATED BY '\n'
40 IGNORE 1 ROWS
41 (id, product_name, @price, colour, weight, warehouse_id, category, brand, @cost, launch_date)
42 SET
43   price = REPLACE(@price, '$', ''),
44   cost = REPLACE(@cost, '$', '');
45
```

The 'Output' pane at the bottom shows the execution results:

#	Time	Action	Message
33	23:22:16	LOAD DATA LOCAL INFILE 'C:/Users/irina/Desktop/DATA/DATA ANALIST I...	100 row(s) affected, 200 warning(s): 1265 I...
34	23:22:22	SELECT * FROM irinasdatabase.products	200 row(s) returned
35	23:22:52	LOAD DATA LOCAL INFILE 'C:/Users/irina/Desktop/DATA/DATA ANALIST I...	100 row(s) affected, 200 warning(s): 1265 I...
36	23:23:06	SELECT * FROM irinasdatabase.products	300 row(s) returned
37	23:38:59	LOAD DATA LOCAL INFILE 'C:/Users/irina/Desktop/DATA/DATA ANALIST I...	100 row(s) affected Records: 100 Deleted
38	23:39:06	SELECT * FROM irinasdatabase.products	100 row(s) returned

2.3 - He utilizado la tabla json para “romper” la serie de 3 códigos de productos de cada campo de la columna “product_ids” en 3 filas x 1 código de producto cada.

Irina_Mihai_Tasca S2.01. Nocio... x companies transactions credit_cards transaction_products

cost = REPLACE(@cost, '\$', '');

```

35484
35485
35486
35487 • SELECT p.product_name, COUNT(*) AS vendes
35488 FROM transaction_products tp
35489 JOIN products p ON tp.product_id = p.id
35490 GROUP BY p.product_name
35491 ORDER BY vendes DESC;
35492
35493 • JOIN JSON_TABLE(
35494   CONCAT('[', product_ids, ']'),
35495   '$[*]' COLUMNS (product_id INT PATH '$')
35496 ) jt
35497
35498

```

Result Grid

	product_name	vendes
▶	Winterfell	10071
	the duel	10059
	dooku solo	10022
	Karstark Dorne	7694
	skywalker ewok	7607

Result 7 x

Output

Action Output

#	Time	Action	Message
✓ 12	21:45:38	SELECT p.product_name, COUNT(*) AS vendes FROM transaction_products tp JOIN products p ON tp...	70 row(s) returned
✗ 13	21:45:39	JOIN JSON_TABLE(CONCAT('[', product_ids, ']'), '\$[*]' COLUMNS (product_id INT PATH '\$'))jt	Error Code: 1064. You have an error in your SQL syntax; check the

```

CREATE TABLE transaction_products (
    transaction_id VARCHAR(50),
    product_id INT,
    PRIMARY KEY (transaction_id, product_id),
    CONSTRAINT fk_tp_transaction
        FOREIGN KEY (transaction_id)
        REFERENCES transaction(id),
    CONSTRAINT fk_tp_product
        FOREIGN KEY (product_id)
        REFERENCES product(id)

```

```
);
```

```
INSERT INTO transaction_products (transaction_id, product_id)
```

```
SELECT
```

```
    t.id,
```

```
    jt.product_id
```

```
FROM transaction AS t
```

```
CROSS JOIN JSON_TABLE(
```

```
    CONCAT('[', REPLACE(t.product_ids, ' ', '), '), ']',
```

```
    '$[*]' COLUMNS (
```

```
        product_id INT PATH '$'
```

```
    )
```

```
) AS jt
```

```
WHERE TRIM(t.product_ids) <> '';
```

```
SELECT
```

```
    p.id AS ID,
```

```
    p.product_name AS Producto,
```

```
    COUNT(tp.transaction_id) AS ventas
```

```
FROM product AS p
```

```
JOIN transaction_products AS tp
```

```
    ON p.id = tp.product_id
```

```
GROUP BY ID
```

```
ORDER BY ventas DESC;
```

